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# HELMHOLTZ RESONATORS: ONE-DIMENSIONAL LIMIT FOR SMALL CAVITY LENGTH-TO-DIAMETER RATIOS

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## 1. INTRODUCTION

The Helmholtz resonator has been studied extensively, primarily due to its use in a large number of practical applications. As a branch element in piping systems, this well-known device consists of a volume communicating with a disturbance in the main duct through an orifice or neck. For the typical low frequency resonator, wave propagation in the main duct and connector may be considered to be planar. At the interface of the volume and connector, however, the discontinuity in cross-sectional area induces multi-dimensional wave propagation in the volume. Since the specific connector and volume dimensions govern the production and dissipation of higher order modes, these parameters have a significant effect upon the resonator performance. Therefore, in order to predict resonator response accurately, a multi-dimensional analysis of the connector–volume interface is usually required [1, 2]. Results of these analyses are often incorporated into the classic lumped parameter Helmholtz resonator model by calculating an effective "end correction" and adding it to the connector length [3].

To obtain a multi-dimensional solution or, equivalently, an accurate estimate of the end correction, an analysis generally requires the solution of a number of simultaneous equations, or a computational approach. However, under certain conditions closed form solutions may be found [3–6]. The present study pursues one of these special cases that appears to have been neglected thus far. By considering concentric Helmholtz resonators with small cavity length-to-diameter ratios, a one-dimensional closed form solution for transmission loss and resonance frequencies is developed. Thus the study investigates the limit as the volume approaches a "pancake" configuration.

## 2. LIMIT FOR $l/d \ll 1$

The geometry considered in the study is shown in Figure 1, consisting of an axisymmetric connector and volume, which are attached to an anechoically terminated main duct. Planar propagation is assumed in the main duct and connector, which requires $kd_p \ll 1$ and $kd_c \ll 1$, where $k = 2\pi/\lambda$ is the wavenumber, $\lambda$ being the wavelength, and $d_p$ and $d_c$ are diameters of the main duct and connector, respectively. Depending on component dimensions and the wavelength, the non-planar effects generated at the connector–volume interface may continue to propagate in the volume or may dissipate away from the



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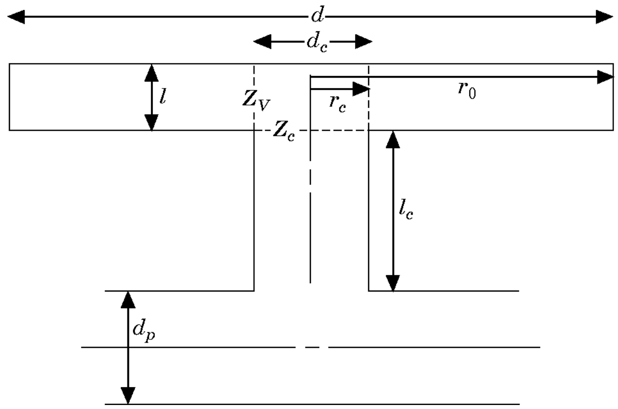


Figure 1. The geometry and nomenclature for the derivation.

discontinuity. If substantial multi-dimensional behavior is generated locally, it will have a significant effect on the resonator response, regardless of whether the higher order modes propagate throughout the entire volume. For the resonator geometry of the present study, higher order mode generation will clearly diminish as $d$ approaches $d_c$. A closed form solution may then be formulated by considering one-dimensional axial wave propagation in the cavity [4–6]. For a low frequency resonator of fixed volume, the effects of non-planar wave propagation are expected to increase as $l/d$ is reduced toward and below unity, thereby necessitating a multi-dimensional analysis. However, as the $l/d$ ratio is decreased further, such that the volume approaches a "pancake" configuration, axial gradients in the volume become negligible, and an additional one-dimensional solution may be obtained. This case, in which purely radial propagation in the volume is considered, is described next.

### 2.1. *Radial propagation*

In the limit for $l/d \ll 1$, wave motion in the volume is approximated as inviscid, one-dimensional radial propagation in a disk of constant thickness. For the analysis, the linearized wave equation,

$$\nabla^2 p = \frac{1}{c_0^2}\frac{\partial^2 p}{\partial t^2}, \tag{1}$$

is expressed in cylindrical co-ordinates as

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{1}{c_0^2}\frac{\partial^2 p}{\partial t^2}, \tag{2}$$

where $p$ is the acoustic pressure, $c_0$ is the mean speed of sound, $r$ is the radius and $t$ is time. Assuming harmonic time dependence for the acoustic variables, the pressure may be written as

$$p = R\mathrm{e}^{\mathrm{i}\omega t}, \tag{3}$$

where $R$ is a function of radius, $\omega = 2\pi f$ is the angular frequency, $f$ being the frequency, and i is the imaginary unit. Inserting this form into equation (2) leads to

$$r^2\frac{\mathrm{d}^2 R}{\mathrm{d}r^2} + r\frac{\mathrm{d}R}{\mathrm{d}r} + k^2 r^2 R = 0, \tag{4}$$

where $k = \omega/c_0$ is the wavenumber. Equation (4) is Bessel's equation of order zero, which may be solved in terms of Hankel functions as

$$R = C_1\, \mathrm{H}_0^{(1)}(kr) + C_2\, \mathrm{H}_0^{(2)}(kr), \tag{5}$$

where a subscripted $C$ represents a complex constant and $\mathrm{H}_j^{(n)}$ is the Hankel function of order $j$ and type $n$. The Hankel functions are given by

$$\mathrm{H}_j^{(1)}(kr) = \mathrm{J}_j(kr) + \mathrm{i}\mathrm{Y}_j(kr), \qquad \mathrm{H}_j^{(2)}(kr) = \mathrm{J}_j(kr) - \mathrm{i}\mathrm{Y}_j(kr), \tag{6, 7}$$

where $\mathrm{J}_j$ and $\mathrm{Y}_j$ are Bessel functions of the first and second kind of order $j$, respectively. The combination of equations (3) and (5) yields the pressure as

$$p = [C_1\, \mathrm{H}_0^{(1)}(kr) + C_2\, \mathrm{H}_0^{(2)}(kr)]e^{\mathrm{i}\omega t}. \tag{8}$$

To determine the velocity fluctuations, consider the linearized Euler equation,

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho_0}\nabla p, \tag{9}$$

or, for purely radial propagation,

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho_0}\frac{\partial p}{\partial r}, \tag{10}$$

where $u$ is the acoustic velocity and $\rho_0$ is the ambient density. In terms of $u = U\mathrm{e}^{\mathrm{i}\omega t}$, where $U$ is a function of $r$, equations (8) and (10) yield

$$u = -\frac{\mathrm{i}}{\rho_0\, c_0}[C_1\, \mathrm{H}_1^{(1)}(kr) + C_2\, \mathrm{H}_1^{(2)}(kr)]e^{\mathrm{i}\omega t}. \tag{11}$$

The boundary condition at the rigid wall in the volume requires $u(r_0, t) = 0$; therefore

$$C_2 = -C_1 \frac{\mathrm{H}_1^{(1)}(kr_0)}{\mathrm{H}_1^{(2)}(kr_0)}. \tag{12}$$

In view of this relation, the expressions for pressure and velocity may be arranged as

$$p = C_1\left[\mathrm{H}_0^{(1)}(kr) - \frac{\mathrm{H}_1^{(1)}(kr_0)}{\mathrm{H}_1^{(2)}(kr_0)}\mathrm{H}_0^{(2)}(kr)\right]\mathrm{e}^{\mathrm{i}\omega t}, \tag{13}$$

$$u = -\frac{\mathrm{i}}{\rho_0\, c_0} C_1\left[\mathrm{H}_1^{(1)}(kr) - \frac{\mathrm{H}_1^{(1)}(kr_0)}{\mathrm{H}_1^{(2)}(kr_0)}\mathrm{H}_1^{(2)}(kr)\right]\mathrm{e}^{\mathrm{i}\omega t}. \tag{14}$$

For notational convenience, an impedance for the volume is defined as

$$Z_V = \left(\frac{p}{u}\right)_{r=r_c} = \mathrm{i}\rho_0\, c_0 \left\{\frac{\mathrm{H}_0^{(1)}(kr_c) - [\mathrm{H}_1^{(1)}(kr_0)/\mathrm{H}_1^{(2)}(kr_0)]\mathrm{H}_0^{(2)}(kr_c)}{\mathrm{H}_1^{(1)}(kr_c) - [\mathrm{H}_1^{(1)}(kr_0)/\mathrm{H}_1^{(2)}(kr_0)]\mathrm{H}_1^{(2)}(kr_c)}\right\}, \tag{15}$$

where the subscript $V$ indicates the location within the volume at $r = r_c$. In the next section this volume impedance is incorporated in expressions for the transmission loss and resonance frequencies of the Helmholtz resonator.

### 2.2. *Transmission loss and resonance frequencies for $l/d \ll 1$*

For one-dimensional wave motion in the main duct, the transmission loss of an acoustic element may be expressed as

$$TL = 10 \log_{10} \left| \frac{C_{+,i}}{C_{+,tr}} \right|^2, \tag{16}$$

where $C_{+,i}$ and $C_{+,tr}$ are complex constants representing the magnitudes of the (harmonic) incident and transmitted pressure waves, respectively. Assuming constant pressures and conservation of volume flow at the main duct–connector intersection and neglecting viscous effects, one-dimensional acoustic theory yields the Helmholtz resonator transmission loss as

$$TL = 10 \log_{10} \left| 1 + \frac{A_c}{2A_p} \left[ \frac{1 + \mathrm{i}(Z_c / \rho_0 c_0) \tan kl_c}{\mathrm{i} \tan kl_c + (Z_c / \rho_0 c_0)} \right] \right|^2, \tag{17}$$

where the impedance at the volume end of the connector has been introduced as

$$Z_c = (p/u)_{connector,\ volume\ end}. \tag{18}$$

Note that the current study concentrates on the volume side of the connector, so multi-dimensional effects at the main duct–connector junction will not be considered explicitly, although they may be incorporated in the effective connector length $l_c$. Inspection of equation (17) reveals that the transmission loss becomes infinite for

$$\mathrm{i} \tan kl_c + Z_c/\rho_0 c_0 = 0, \tag{19}$$

which defines the resonance frequencies.

For the overall resonator treatment, equations (17) and (19) are general, provided that $Z_c$ may be obtained for the volume. To determine $Z_c$ in the limiting case for $l/d \ll 1$, the region in the volume from $r = 0$ to $r_c$, denoted by subscript $J$, is taken as a constant pressure junction; therefore,

$$p_c = p_J = p_V, \tag{20}$$

where subscript $c$ denotes the location at the volume end of the connector. Assuming that the junction volume is compressed adiabatically,

$$p_J = p_c = -\rho_0 c_0^2 \frac{\mathrm{d}V_J}{V_J} = \rho_0 c_0^2 \frac{\xi_c A_c - \xi_V A_V}{V_J}, \tag{21}$$

where $\xi$ is the acoustic displacement, $A_V = 2\pi r_c l$ is the curtain area in the volume at $r = r_c$ and $V_J = \pi r_c^2 l$ is the junction volume. For both planar and radial propagation the velocity and particle displacement are related by $u = \mathrm{i}\omega\xi$, which allows equation (21) to be written as

$$p_c = \frac{\rho_0 c_0^2}{\mathrm{i}\omega V_J} (u_c A_c - u_V A_V). \tag{22}$$

In terms of $Z_V$,

$$u_V = p_V / Z_V = p_c / Z_V, \tag{23}$$

and equation (22) may then be rearranged to yield

$$Z_c = \frac{p_c}{u_c} = \frac{1}{\mathrm{i}kV_J / \rho_0 c_0 A_c + A_V / Z_V A_c}, \tag{24}$$

or

$$Z_c = \frac{1}{\mathrm{i}kl/\rho_0 c_0 + 2l/Z_V r_c}. \tag{25}$$

Substituting equation (25) into equations (17) and (19) yields the final expressions for the transmission loss and resonance frequencies, respectively. If one-dimensional propagation in the radial direction applies, $kl \ll 1$, and the first term in the denominator of equation (25) is, most likely, inconsequential. Neglecting this term is the same as considering $V_J$ to be incompressible, yielding a slightly simpler expression for $Z_c$ as

$$Z_c = Z_V (r_c/2l). \tag{26}$$

However, in the limit as $r_0$ approaches $r_c$, $Z_V$ becomes infinite, and equation (26) predicts performance equivalent to a quarter-wave resonator of length $l_c$. The benefit of using the original expression, equation (25), is that it reduces to the lumped parameter result,

$$Z_c = -\mathrm{i}\rho_0 c_0 \frac{A_c}{kV}. \tag{27}$$

as $r_0$ approaches $r_c$.

## 3. RESULTS

The most significant effect of wave motion in the resonator volume is to change the location of the primary resonance frequency. Inspection of equations (19) and (25) also reveals that additional length-controlled resonances will occur which are not accounted for in the lumped theory. However, for a low frequency resonator, these secondary resonances will usually occur at frequencies above the range of interest. For a resonator having fixed connector geometry and cavity volume, Figure 2 compares the primary resonance frequencies computed from: (1) one-dimensional radial analysis, equation (25); (2) lumped volume approach, equation (27); and (3) the one-dimensional axial solution, where $Z_c = -\mathrm{i}\rho_0 c_0 (r_c/r_0)^2 \cot kl$ [4–6]. In order to include a wide range in resonator geometry, the $l/d$ values are presented on a logarithmic scale. A reduction in the primary resonance frequency for low values of $l/d$ is clearly shown in Figure 2. This trend for circular concentric resonators agrees qualitatively with the results of Chanaud [2] for cavities of rectangular cross-section. As $l/d$ is increased, approaching the order of

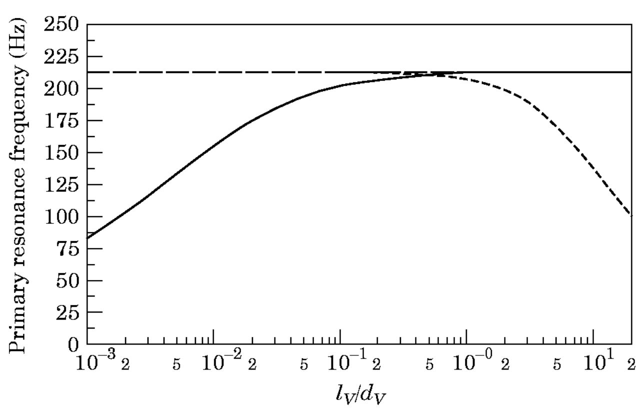


Figure 2. The primary resonance frequency of concentric Helmholtz resonator from 1-D radial (——), 1-D axial (– – –), and lumped volume (——), analyses. $V = 1000$ cm$^3$, $r_c = 2{\cdot}0$ cm, $l_c = 8{\cdot}0$ cm.

unity, all three methods tend to yield similar results. With further increase in $l/d$, axial propagation begins to dominate. It should be noted that as the volume length becomes very small, the effects of viscosity which are neglected in the present study are expected to become significant. Although an unsteady solution for viscous radial flow in a disk does not appear to be available, some insight may be gained from the work of Tidjeman [7] or Mikhail and El-Tantawy [8], who considered oscillating viscous flow in cylindrical tubes and two-dimensional slots, respectively. The effect of viscous dissipation is currently being investigated and the results will be reported elsewhere.

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